

1. Real-World Software Problem

The system is organized into a tree-like hierarchy:

- **First Floor:** Includes the Living Room (Light, Air Conditioner, Smart TV) and the Kitchen (Light, Smart Oven).
- **Second Floor:** Includes the Bedroom (Light, Heater) and a Security Area (Camera, Smart Door Lock).

The primary goal is to provide seamless control at any level—whether interacting with a single device, an entire room, or a whole floor.